WHERE DOES YOUR DATA ACTUALLY LIVE?

DATA LAKE

VS

DATA WAREHOUSE

VS

LAKEHOUSE

Which one should a Data Engineer use?

RAW DATA

ANALYTICS

AI/ML

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WHAT IS A DATA LAKE?

A Data Lake is a centralized repository that stores large amounts of data in its raw, native format.

DATA CAN COME FROM ANYWHERE

- Applications
- Databases
- APIs
- Files (CSV, JSON, XML)
- Logs & Events

CAN HOLD

- ✓ Structured Data
- ✓ Semi-structured Data
- ✓ Unstructured Data
- ✓ Raw & Processed Data

SUPPORTED FORMATS

- CSV
- JSON
- Parquet
- Avro
- ORC
- Images, Videos
- Logs and more...

Think of it as a large lake where all kinds of data flows in and is stored for future use.

COMMON TECHNOLOGIES

Amazon S3	Azure Data Lake Storage	Google Cloud Storage	On-premise HDFS

EXAMPLE

An e-commerce company stores orders, clicks, payments, logs, images and events in a Data Lake.

FLEXIBLE STORAGE TODAY, ENDLESS POSSIBILITIES TOMORROW!

WHAT IS A DATA WAREHOUSE?

A Data Warehouse is a centralized repository of cleaned, integrated and structured data optimized for analytics and business intelligence.

TYPICAL DATA WAREHOUSE FLOW

TA SOURCES
s, DB, APIs, Files)

→ ETL / ELT
(Extract, Transform, Load)

→ DATA WAREHOUSE
(Curated & Modeled Data)

→ SQL QUERIES & ANALYTICS

→ BI TOOLS & DASHBOARDS
(Reports, Insights)

KEY CHARACTERISTICS

- Stores curated and structured data
- Schema-on-write (Highly structured)
- Optimized for fast SQL queries
- Strong data quality & consistency
- Designed for reporting and BI

COMMON USE CASES

- Business reporting & dashboards
- Sales & revenue analytics
- Customer analytics
- Financial reporting
- KPI tracking

EXAMPLES OF DATA WAREHOUSE TECHNOLOGIES

Snowflake | Amazon Redshift | Google BigQuery | Azure Synapse Analytics | Others...

REMEMBER:

Data Lake stores EVERYTHING.
Data Warehouse stores what is IMPORTANT for analytics.

Think of it as a highly organized store built for decision making!

WHAT IS A LAKEHOUSE?

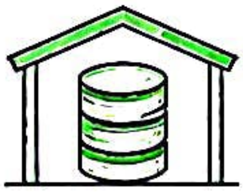
A Lakehouse combines the best of Data Lake and Data Warehouse. It brings warehouse-style management and performance to data stored in a lake-style architecture.

DATA LAKE STORAGE + WAREHOUSE CAPABILITIES = LAKEHOUSE

DATA LAKE STORAGE

Scalable, flexible, stores all types of data

+ WAREHOUSE CAPABILITIES



- ACID Transactions
- Data Governance
- Schema Enforcement
- Data Quality
- Indexing & Optimization

= LAKEHOUSE

Unified platform for all data workloads

SUPPORTS MULTIPLE WORKLOADS

DATA ENGINEERING
Build, transform and process data

SQL ANALYTICS
Fast queries on large datasets

BI & REPORTING
Dashboards and business insights

DATA SCIENCE
Explore, analyze and experiment

MACHINE LEARNING
Train and serve ML models

COMMON LAKEHOUSE TECHNOLOGIES

DELTA LAKE
Reliable data lakes with ACID transactions

APACHE ICEBERG
Open table format for huge analytic datasets

APACHE HUDI
Incremental processing & real-time pipelines

One data foundation, endless possibilities!

LAKEHOUSE = THE FUTURE OF MODERN DATA PLATFORMS

DATA LAKE VS DATA WAREHOUSE VS LAKEHOUSE

FEATURE	DATA LAKE	DATA WAREHOUSE	LAKEHOUSE
MAIN PURPOSE	Flexible storage of raw data	Optimized for analytics & BI	Unified analytics, ML & data workloads
DATA STORED	Raw + Processed	Curated / Modeled	Raw + Curated
DATA TYPES	Structured, Semi-structured, Unstructured	Structured / Tabular	Structured, Semi-structured, Unstructured
SCHEMA APPROACH	Schema-on-Read	Schema-on-Write	Schema-on-Read & Schema-on-Write
PERFORMANCE (ANALYTICS)	Needs extra processing layers	High performance for SQL queries	High performance with open formats
BEST SUITED FOR	Data Science, ML, Exploration	BI, Reporting, Business Analytics	BI, ML, Data Science, Data Engineering
STORAGE	Object Storage (S3, ADLS, GCS)	Warehouse managed storage	Object Storage + Table Management
FLEXIBILITY	Very High	Moderate	High + Governed
COST	Lower storage cost	Higher compute cost	Cost optimized with open formats
EXAMPLES	Amazon S3, ADLS, Google Cloud Storage	Snowflake, Redshift, BigQuery, Synapse	Delta Lake, Iceberg, Hudi on S3/ADLS/GCS

EASY WAY TO REMEMBER

DATA LAKE

STORE EVERYTHING

→

DATA WAREHOUSE

ANALYZE IMPORTANT DATA

→

LAKEHOUSE

UNIFY & POWER ALL WORKLOADS

THERE IS NO ONE-SIZE-FITS-ALL. CHOOSE BASED ON YOUR DATA, USE CASES, GOVERNANCE & BUSINESS NEEDS.

DATA LAKE vs DATA WAREHOUSE vs LAKEHOUSE

WHICH ONE SHOULD YOU CHOOSE?

CHOOSE DATA LAKE

- You need to store large volumes of raw, diverse and unstructured data.
- You need maximum flexibility.
- Use cases: Exploration, Data Science, ML, Logs, IoT, Clickstream.

BEST FOR: Storage, Exploration & Future Use Cases

CHOOSE DATA WAREHOUSE

- Your primary goal is analytics, reporting and business intelligence.
- You need fast SQL queries and high performance.
- Use cases: Dashboards, Financial Reporting, KPI tracking, Business Analytics.

BEST FOR: Analytics, BI, Reporting

CHOOSE LAKEHOUSE

- You want the best of both worlds.
- You need analytics + ML on the same data.
- You want unified data platform with governance and performance.
- Use cases: BI, ML, Data Engineering, Streaming Analytics.

BEST FOR: Unified Platform for Analytics + ML + DE

INTERVIEW TAKEAWAY

DATA LAKE → Flexible storage for raw & diverse data.

DATA WAREHOUSE → Organized, curated data for fast analytics & BI.

LAKEHOUSE → Lake flexibility with warehouse capabilities.

THINK OF IT AS...

LAKE = STORE

↓

WAREHOUSE = ANALYZE

↓

LAKEHOUSE = UNIFY

THERE IS NO SILVER BULLET!
Understand your DATA, WORKLOADS, GOVERNANCE & BUSINESS GOALS.
THEN CHOOSE WISELY!

SAVE THIS for your next Data Engineering interview!